Vedant Thapar

website | vedant.thapar@gmail.com | linkedin.com/vedant | github.com/RazerBlade6

Computer Science graduate (BITS Pilani Dubai, 2026) building AI-driven automation, API-driven backends, and data pipelines in Python, Go, and Node.js. Currently developing agent orchestration services with LangChain at 1Kosmos; previously automated cloud data pipelines on GCP at Mondelez and built test automation tooling at Underscore Cybersecurity. Hands-on with RAG, MCP tooling, LangGraph, FastAPI, AWS, PostgreSQL, Docker, and Kubernetes on Linux.

Education

BITS Pilani, Dubai Campus — Dubai, UAE
Bachelor of Engineering, Computer Science | CGPA: 8.62 — *Sep. 2022 – Jun. 2026*

Experience

Intern, R&D — Aug 2026 – Present
1Kosmos Cybersecurity — *New Delhi, India*

- Designing and developing SaaS APIs for CRUD agents using Node.js and Python, standardizing agent interactions within the company's internal agentic services platform
- Creating an agent control plane using Python and LangChain for centralized agent creation, orchestration, and lifecycle management.

Strategy Intern — Aug 2025 – Jan 2026
Mondelez International — *Dubai, UAE*

- Built automated data pipelines integrating legacy Sales and RTM reporting systems with Google Cloud Platform analytics infrastructure, based on BigQuery with Spark
- Developed Python scripts to scrape and process NielsenIQ product datasets, improving internal reporting automation
- Collaborated with the Global Head of Digital Services on AI validation initiatives aimed at detecting fraudulent Mondelez products on retail shelves
- Worked with the Digital Acceleration team to design reporting tools supporting promotion planning and performance tracking

Cybersecurity Intern — May 2024 - Aug 2024
Underscore Cybersecurity — *New Delhi, India*

- Developed 'F-Detector', a Python-based network analysis tool using Fourier Transform techniques to detect bot-like activity, improving detection accuracy by 30%
- Built an internal Python framework with a team of six engineers to streamline cybersecurity testing workflows
- Enhanced internal testing tools, increasing automated testing efficiency by 20%

Web and SEO Intern — Jun 2025 - Aug 2025
Innotek – Kosha Yoga — *Mumbai, India*

- Managed and optimized the company website, ensuring 99% uptime across mobile and desktop environments.
- Implemented technical SEO improvements including metadata structuring, keyword optimization, and sitemap generation, increasing organic traffic by 30%
- Configured hosting infrastructure, DNS systems, and server security protocols for reliable and scalable deployment
- Developed a Python-based chatbot that improved customer engagement and reduced response time for user queries

Projects

Apex | *Go, SQLite*

- Go terminal application that models a developer's skills and project portfolio and uses it to generate prioritized action items, propose new projects, and implement them on request
- Two separate loops: an LLM advisor reasoning across the whole portfolio, and a builder that dispatches briefed tasks to an external coding agent behind a swappable interface, streaming structured events into a Bubble Tea TUI.
- Provider abstraction spanning the Anthropic and OpenAI APIs plus subscription-backed inference through the Claude Code CLI, normalizing streaming and schema-constrained output behind a single interface.

- Local-first pure-Go SQLite persistence with forward-only additive migrations, compare-and-swap cross-machine locking and flock-based concurrency control, keeping the full portfolio in-context via per-project digests keyed to a content-and-git-state hash.

Engram | *Next.js, FastAPI, PostgreSQL (pgvector), Supabase, Docling, Anthropic API*

- Full-stack study app that ingests textbooks, slides and notes, then generates flashcards, cloze and multiple-choice cards scheduled with the FSRS algorithm. Every card and chat answer links to the exact source page, with the passage highlighted.
- Provider-agnostic LLM layer that routes each job to a model based on benchmark results and uses Claude native Citations. Added CI evaluation suites that measure retrieval quality (recall@k, MRR), card faithfulness with an LLM judge calibrated against user flags, and agreement between the LLM grader and human graders
- Layout-aware ingestion pipeline using Docling and vision-model figure captioning, with section-aware chunking using a custom hybrid retrieval system (pgvector plus full-text search), in-process ONNX embeddings, and long ingestion jobs on a worker backed by a Postgres job queue

SQLSense | *LangGraph, AWS (Bedrock, Athena, Glue, Lambda, Aurora Serverless + pgvector, DynamoDB, S3)*

- Built a LangGraph multi-agent natural language to SQL system. A planner sends each question to a SQL agent (Athena over a partitioned S3 Parquet lake), a retrieval agent (hybrid pgvector and keyword search merged with RRF), or both. Answers show the SQL, citations, a chart and a per-step agent trace, and DynamoDB session memory handles follow-up questions
- SQL generation kept safe with sqlglot syntax-tree validation (single SELECT, table allowlist, forced LIMIT), read-only IAM, a 100 MB Athena scan cap and a self-correcting retry loop. Added a warehouse interface with Athena and DuckDB backends over the same Parquet files, so tests and local development run offline.
- Developed against an evaluation suite that scores execution accuracy, routing accuracy, retrieval recall@5 and faithfulness, all gated in CI/CD through GitHub Actions OIDC and achieving 94–97% execution accuracy. All infrastructure is defined in AWS CDK.

Technical Skills

Languages: Python, Go, C/C++, Rust, Java, JavaScript, TypeScript, SystemVerilog
Full-Stack: React, Node.js, FastAPI, REST API design, Streamlit
AI Engineering: LangChain, LangGraph, RAG, MCP, Anthropic and OpenAI APIs, PyTorch, TensorFlow, Keras
Data Engineering: PostgreSQL, S3, MongoDB, SQLite, ChromaDB, BigQuery, Apache Spark, Athena, Supabase
Cloud and DevOps: AWS, Google Cloud Platform, Docker, Kubernetes, Git, GitHub, Linux